



TEXT WATERMARKING · PAN TASK 3

Post-Hoc Synonym Watermarking

Dual-Key Type-Level Detection, and the
precision–robustness trade-off it reveals

AI > MØMEnt

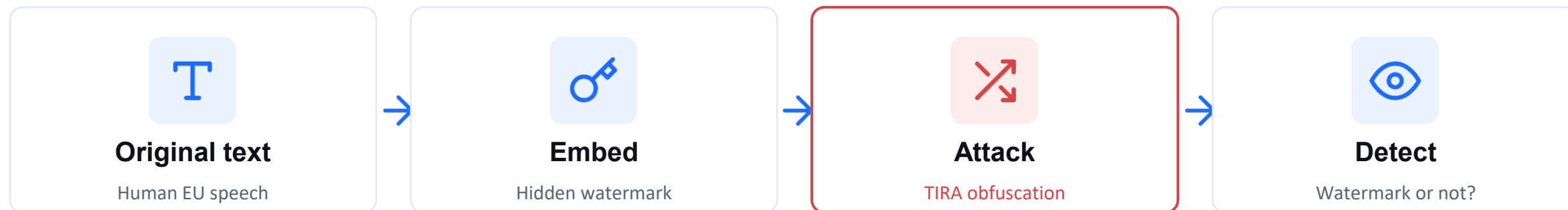
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Rank #2 of 8 teams · Best spot-check TWF 0.971 (dual-key) · Best official TWF 0.775 (single-key) — a trade-off story

Post-hoc watermarking under attack

Embed an invisible, detectable signal into pre-existing human EU-Parliament speeches — not during generation. Survive an automated obfuscation attack, then detect the watermark.



THE METRIC — Text Watermarking Fidelity

$$\text{TWF} = \max(\text{BLEU}, \text{BERTScore}) \times \text{Balanced Accuracy}$$

Fidelity — how little the text changed × *Detection — did we get it right, penalising misses & false alarms equally*

A three-way tension — sharpened by attack.

Post-hoc watermarking must satisfy three goals at once — and an obfuscation attack that removes part of the signal makes the balance harder.



High TP

Detect the watermark reliably, even after attack



Near-zero FP

Never flag an un-watermarked original



Semantic fidelity

Preserve the original meaning & wording

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How It Works

Cryptographic partitioning, synonym substitution, type-level detection

Green / red vocabulary via HMAC

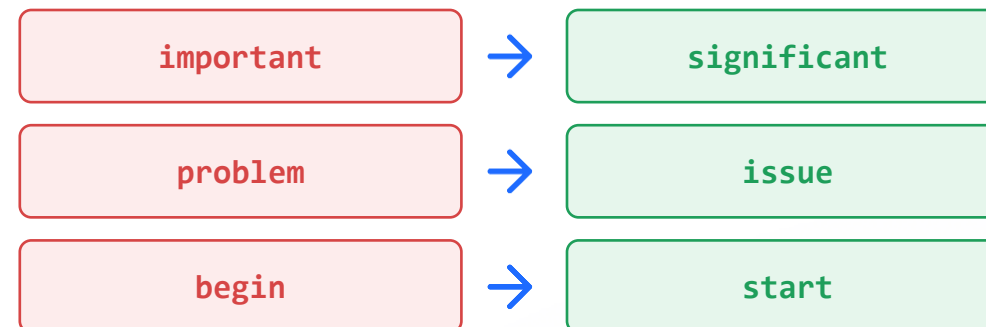
A secret key deterministically splits the English vocabulary into green and red word types. We globally replace red words with green synonyms — pushing the green fraction up in a detectable, key-specific way.

$$\text{bit}(w; K) = \text{HMAC-SHA256}(K, w) \bmod 2$$

$\text{bit} = 1 \rightarrow \text{green}$ · $\text{bit} = 0 \rightarrow \text{red}$

Same word + same key $\rightarrow$ same color; different key $\rightarrow$ independent partition

Example substitution



Two design choices set this apart:



Type-level detection — count each unique word once, not per occurrence.



Dual-key AND detection — require two independent keys to pass.

Type-level detection

The detector runs a one-sided Z-test on the green fraction. Crucially, we count unique word types — each identity once — rather than token occurrences.



Token-level (the problem)

- > Repetitive political terms (parliament, committee, member) appear dozens of times.
- > One substitution of such a word dominates the Z-statistic.
- > The null distribution becomes unstable & skewed.



Type-level (our fix)

- > Each word identity counts exactly once, regardless of repetition.
- > Every substitution gets equal statistical weight.
- > A stable, well-centred null — ideal for repetitive EU-speech vocabulary.

Expanding POS coverage {NN, JJ} → {NN, NNS, JJ, VB, VBP} raised avg. scoreable types 11.6 → 24.6 (28 failure cases → 0 failure cases), eliminating all 28 undetectable texts.

> DESIGN CHOICE 2 · THE KEY IDEA

Dual-key AND detection

Classify as watermarked only if BOTH independent Z-tests pass. Two chances to raise a false alarm become one joint, far rarer event.

$$Z(K1) \geq 1.5 \quad \text{AND} \quad Z(K2) \geq 1.5$$

Single key

≈6.7%

P(false positive)



Dual key (AND)

≈0.45%

P(false positive)

The catch

- > P(FP) drops 15× — extremely precise.
- > But AND fails if the attack suppresses EITHER key below threshold.
- > One weak signal sinks the whole detection — even if the other survives strongly.

Three sequential stages



Stage 1 — Key-1

Select top-12 red types (frequency-first), replace globally with green synonyms. Prefer double-green synonyms to pre-boost Key-2.



Stage 2 — Key-2

Repeat for the second key, skipping types already used. May lower the Key-1 signal.



Stage 3 — Restoration

Up to 3 rounds alternate between keys to a conservative target, restoring both Z-scores above threshold.

Robustness safeguards built in



Frequency-first ranking

High-frequency types are harder to erase → embedded first



Text hardening

NFKC normalisation, homoglyph remap, zero-width removal



Semantic fallback

BERT fill-mask when WordNet lacks a green synonym (SBERT sim ≥ 0.45)

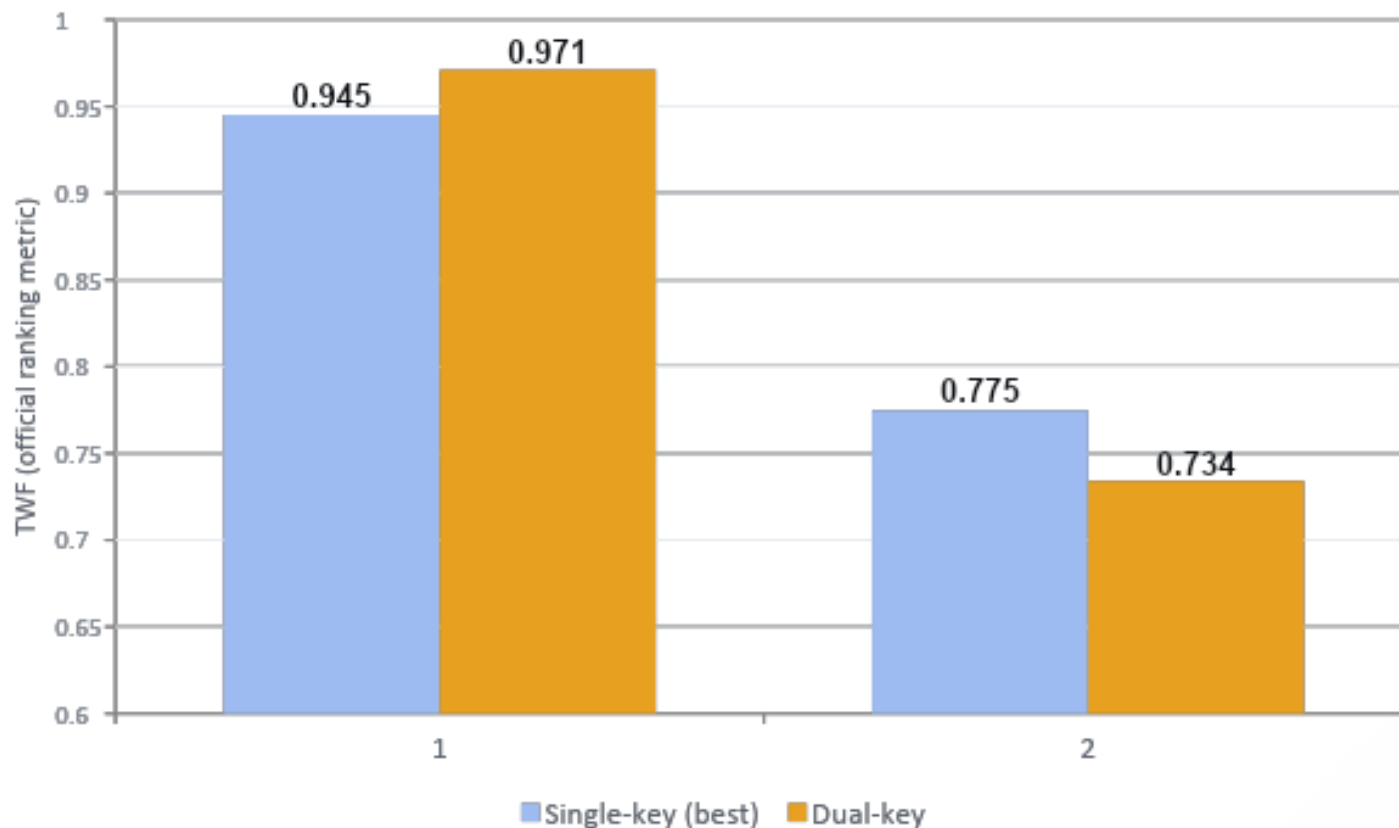
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Results & The Trade-off

Where dual-key wins, where single-key wins

A stark inversion between attacks



Read the crossover

- > Mild attack: dual-key wins (0.971) — both signals survive, precision pays off.
- > Strong attack: single-key wins (0.775) — dual-key TP collapses 0.496 → 0.254.
- > The same design flips from best to worst as the attack hardens.

The precision–robustness trade-off

The mechanism

- > Shuffling is position-based; our $\text{bit}(w;K)$ depends only on word identity → zero signal loss, AND holds easily.
- > The strong attack substitutes words, removing embedded green types.
- > With margin 1.0 above threshold, the attack still suppresses ≥ 1 key in $\sim$ half of texts.
- > AND then fails — one downed key sinks detection.

A structural law

$$P(\text{detect}) = \prod P(\text{key } i \text{ survives})$$

Each added key multiplies in another survival probability < 1 . Under uniform attack degradation, more keys strictly lowers TP.

Dual-key TN stayed 0.499 on BOTH datasets — it never false-detects. Precision is bought at the cost of missed detections once both signals cannot survive together.



Zero false positives, always — the precision promise held.

A design space, not a winner

The lesson generalises to any multi-key AND scheme. The choice of detection logic should match the expected attack strength.



AND condition

Two keys must pass

Lowest FP (0.45%) · brittle under strong attack

Best for mild / no attack



OR condition

Either key suffices

More robust · higher FP (≈12.9%)

Best when robustness dominates



Hybrid

AND at lower threshold, or switch to OR on a very high Z

Recovers detection without losing all precision

A promising middle ground

> CONCLUSION

What to take away



Two design choices Type-level Z-test detection + a dual-key AND criterion, over cryptographic HMAC synonym substitution.



A characterised trade-off Dual-key cuts FP 15× (6.7% → 0.45%) but is brittle: TWF 0.971 spot-check vs 0.734 official.



Single-key won official Best official TWF 0.775 — only one signal must survive, so it is structurally more robust.



FUTURE WORK

- > Hybrid AND/OR detection logic
- > Higher embedding targets (3.0–3.5) for more margin
- > Richer synonym sources beyond WordNet + BERT-LS



github.com/Momen-Ibrahim-Fawzy/posthoc-dualkey-watermarking--PAN2026

Thank You

Questions welcome

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